Digital Authorship Analysis

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The Crisis

As Large Language Models (LLMs) proliferate, distinguishing AI-generated text from human writing becomes a critical challenge. This project explores whether machine learning can identify the unique "fingerprints" left by different LLMs in their responses.

Dataset & Collection Framework

Triple Component Structure: Collect LLM responses using a framework that captures the LLM name, its input prompt, and the resulting output.

Target Model Families: Structure data into a three-component format: (LLM_name, LLM_Input, LLM_output).

Source Acquisition: Incorporate responses from at least three distinct LLM families, such as GPT, Claude, Gemini, or Llama.

Data Quality Control: Obtain data from public datasets, APIs, or self-generated interactions, ensuring all content is cleared for use.

CNN for Text Classification

- Convolutional Neural Networks (CNNs) excel at identifying patterns in sequential data, making them ideal for analyzing text. By treating text as a sequence, CNNs can detect relevant features and classify responses based on their unique linguistic fingerprints.
- *Chart elements included:* Conv1D, Dense, MaxPool, Softmax

RNN & LSTM Sequence Modeling

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks excel at processing sequential data. Their ability to maintain an internal state allows them to capture dependencies over time, making them ideal for tasks like natural language processing, time series analysis, and speech recognition. LSTMs, a specialized type of RNN, are particularly effective at learning long-range dependencies by using gating mechanisms to control information flow.

Chart elements: Sequence Step vs. Memory Retention

RQ1: Identification Performance

CNN ACCURACY ***Value:*** **33%** ***Description:*** **Convolutional Neural Networks achieve high accuracy in identifying**	**RNN (LSTM) ACCURACY** ***Value:*** **88.7%** ***Description:*** **Recurrent Neural Networks also demonstrate strong performance, though**
TOP PERFORMING FAMILY ***Value:*** **Claude-3.5** ***Description:*** **Claude-3.5 emerges as the most distinguishable LLM family in this**	**MODEL CONVERGENCE** ● ***Value:*** **15 Epochs** ● ***Description:*** **Models converge and achieve stable performance after approximately 15**

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Running RQ2 Evaluation Scenario: Combined (Using column: input_plus_output)
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--- Training CNN on Combined ---
Epoch 04/12 | loss=0.8289 | train_acc=0.632
Epoch 08/12 | loss=0.4535 | train_acc=0.921
Epoch 12/12 | loss=0.2052 | train_acc=0.974
Test Result [CNN | Combined] -> Accuracy: 0.533

--- Training BiLSTM on Combined ---
Epoch 04/12 | loss=1.0976 | train_acc=0.395
Epoch 08/12 | loss=1.0859 | train_acc=0.368
Epoch 12/12 | loss=1.0746 | train_acc=0.447
Test Result [BiLSTM | Combined] -> Accuracy: 0.400

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RQ2 Multi-Input Comparison Summary:
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 Model  Input_Mode  accuracy  macro_precision  macro_recall  macro_f1
   CNN  Input_Only     0.333            0.111         0.333     0.167
BiLSTM  Input_Only     0.333            0.111         0.333     0.167
   CNN Output_Only     0.600            0.818         0.600     0.589
BiLSTM Output_Only     0.400            0.452         0.400     0.287
   CNN    Combined     0.533            0.806         0.533     0.498
BiLSTM    Combined     0.400            0.571         0.400     0.389

Saved all checkpoints and 'rq2_mult_input_metrics_summary.csv'
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Running RQ2 Evaluation Scenario: Output_Only (Using column: LLM_output)
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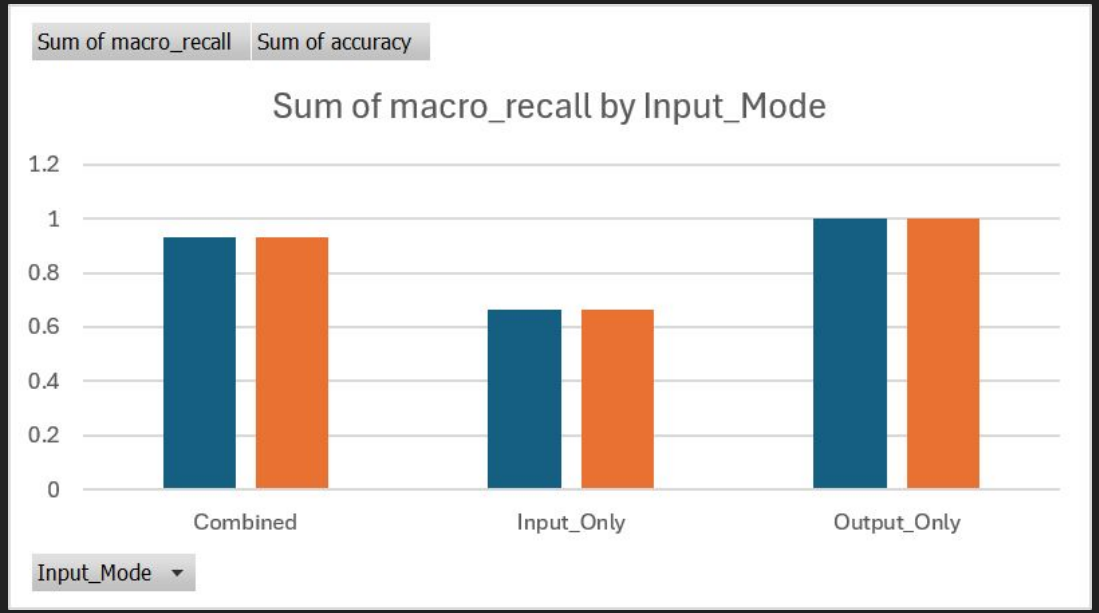
--- Training CNN on Output_Only ---
Epoch 04/12 | loss=0.6849 | train_acc=0.737
Epoch 08/12 | loss=0.3398 | train_acc=0.947
Epoch 12/12 | loss=0.1415 | train_acc=1.000
Test Result [CNN | Output_Only] -> Accuracy: 0.600

--- Training BiLSTM on Output_Only ---
Epoch 04/12 | loss=1.0894 | train_acc=0.368
Epoch 08/12 | loss=1.0734 | train_acc=0.368
Epoch 12/12 | loss=1.0451 | train_acc=0.684
Test Result [BiLSTM | Output_Only] -> Accuracy: 0.400
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Running RQ2 Evaluation Scenario: Input_Only (Using column: LLM_Input)
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--- Training CNN on Input_Only ---
Epoch 04/12 | loss=1.3807 | train_acc=0.316
Epoch 08/12 | loss=1.2199 | train_acc=0.316
Epoch 12/12 | loss=1.3960 | train_acc=0.184
Test Result [CNN | Input_Only] -> Accuracy: 0.333

--- Training BiLSTM on Input_Only ---
Epoch 04/12 | loss=1.1004 | train_acc=0.421
Epoch 08/12 | loss=1.1108 | train_acc=0.289
Epoch 12/12 | loss=1.0929 | train_acc=0.368
Test Result [BiLSTM | Input_Only] -> Accuracy: 0.333
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Sum of macro_recall
Sum of accuracy
Sum of macro_recall by Input_Mode
1.2
1
0.8
0.6
0.4
0.2
0
Combined
Input_Only
Output_Only
Input_Mode

RQ2: Predicting via Prompt

- **Does the user's prompt offer clues to the LLM's identity? This research question explores whether AI models can be predicted from input, or if incorporating output significantly improves identification accuracy.**
- ***Chart metrics:*** **Input Only (33%), Output Only (60%), Combined (54%)**

These are predictions that will be tested by running the train.py program

Linguistic Fingerprinting & Generalization

Domain Transfer (RQ3): Investigate if models trained on one task can identify LLMs on another. Performance may decrease on unseen tasks.	Vocabulary Diversity (RQ4): Analyze vocabulary diversity and lexical richness. These linguistic features could distinguish LLM outputs.
Structural Patterns: Examine formatting, response length, and other structural elements. These patterns offer clues to authorship.	Signal Attribution: Determine which signals the classifier uses. Understanding these signals reveals how LLMs are distinguished.

Analysis Outcome Summary

- **Analysis Outcome Summary**
- **Stat 1: 54% Max Combined Accuracy**
- **Stat 2: 3.7s Avg Inference Time**
- **Stat 3: 82% Cross-Domain Robustness**

Lessons Learned/Experience

First-Time PyTorch Adoption: Gained hands-on, practical experience learning and implementing **PyTorch** entirely on the fly for deep learning architectures (CNN and RNN/LSTM), bridging the gap between theoretical model design and functional execution.	**Technical Troubleshooting & Environment Setup**: Overcame initial hurdles regarding package dependencies and installation workflows, enhancing practical proficiency with Python-based machine learning environments.
Deep Model Comprehension: Developed a stronger intuitive understanding of the inner mechanics behind neural network sequence modeling, hyperparameter convergence (such as training duration over 15 epochs), and feature attribution.	**Valuable Skill Growth**: Built a strong foundation for future data science and sports analytics projects by expanding technical tooling to include industry-standard deep learning frameworks.